Part A:

Question 1:
C

Question 2:
B

Question 3:
A

Question 4:
B

Question 5:

Part B

Question 6:
“6 14”

Question 7:
“15”

Question 8:
“6
9”

Part C

Question 9:
The stack memory allocation stores local functions/variables in the call stack and is managed automatically by the system itself. However, its limited space may also lead to a stack overflow error. Heap memory is allocated dynamically while being handled/managed by the user and is usually much larger. Unlike stack, heap can be resized and altered easily.

Question 10:

A memory leak is when a program uses a portion of the system's memory, but fails to release the space back to said system. It can be prevented by meticulously deleting all unneeded arrays, and using pointers to avoid excessive variable copying.

Question 11:

Const int* p makes an integer P which is a pointer(affects the address/original variable) and cannot have the integer value itself be changed. *p = 5 Should not be written, because a pointer variable is meant to "point" to an already existing variable and affect that specific address (as in not just a number). p = &other would require two additions: An asterisk "*p" and an already existing variable of "int other = #".

Question 12:

The question specifies that reallocation is triggered when the array is full, so reallocation is triggered at values 2, 4, & 8, totalling to 3 times. The total capacity should end at 16. (Isn't this the same as question 4?)